

Module 05: Action in High School

Learning Objectives

1. Define **Action** within the context of High School.
2. Analyze how Action interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Action.

Introduction

This module explores **Action**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Action is a critical component of the 8-part Active Inference spine, bridging the gap between Cognition and Learning.

Key Concepts

1. Action as a Markov Blanket Boundary

How does Action define the boundary between the agent and the environment?

2. Generative Models of Action

What parameters involved in Action must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Action drive the perception-action loop?

Applications

In High School, we see Action manifest in: * **Specific Example 1:** In calculus-based optimization, gradient descent is the mathematical analog of Active Inference action: just as

rolling a ball downhill follows the steepest slope to reach a valley (minimum), an agent selects actions that follow the gradient of free energy downward -- each step reduces the gap between predicted and observed states, and the learning rate parameter controls how aggressively the agent acts on prediction errors. * **Specific Example 2:** The mathematical framework of control theory shows action as solving a differential equation: a thermostat's PID controller computes the error between desired temperature (setpoint) and measured temperature, applies proportional, integral, and derivative corrections, and outputs a heating action -- this closed-loop control is a concrete mathematical implementation of Active Inference's action-as-prediction-fulfillment principle.

Conclusion

Understanding Action allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.